

ECO WARRIORS INITIATIVE UGANDA

Frontend & Backend Website Architecture

Technical blueprint for the public website, administration system, content management, database and media storage.

Core principle

The frontend displays Eco Warriors' information to the public. The backend manages the information, users, projects, activities, stories, images, reports and impact data.

2026

1. System Overview

VISITOR → FRONTEND → API → BACKEND → DATABASE / MEDIA STORAGE

The system should be divided into a public-facing frontend and a secure backend/admin system. The two layers communicate through an API or server-side data layer.

2. FRONTEND — Public Website

The frontend is everything visitors see and interact with.

Page	Route	Purpose
Home	/	Organization introduction, featured programs, projects, impact and calls to action.
About	/about	Story, mission, vision, values, objectives and approach.
Leadership	/about/leadership	Leaders, team, Goodwill Ambassador and relevant profiles.
Programs	/programs	All strategic program areas.
Program Details	/programs/[slug]	Detailed information and projects under a program.
Projects	/projects	All current and past projects.
Project Details	/projects/[slug]	Complete project information and evidence.
Activities	/projects/[slug]/activities	Outreaches, trainings, events and activities.
Stories	/stories	Organization-wide stories of change.
Project Stories	/projects/[slug]/stories	Stories connected to a specific project.
Gallery	/gallery	Organization-wide photo gallery.
Project Gallery	/projects/[slug]/gallery	Project-specific photographs.
Impact	/impact	Impact indicators and results.
Reports	/reports	Annual reports, strategic plans and project reports.
News	/news	Updates, announcements and articles.
Events	/events	Upcoming and past events.
Partners	/partners	Partners, supporters and collaboration opportunities.
Get Involved	/get-involved	Volunteer, partner and advocacy opportunities.
Donate	/donate	Donation and support information.
Contact	/contact	Contact details and enquiry form.

3. Frontend Component Structure

Home
■■■ Navbar
■■■ HeroSection
■■■ WhoWeAre
■■■ ProgramCards
■■■ FeaturedProjects
■■■ ImpactStatistics
■■■ FeaturedStory
■■■ LatestActivities
■■■ PartnerLogos
■■■ CallToAction
■■■ Footer

```
ProjectPage
■■■ ProjectHero
■■■ ProjectOverview
■■■ ProjectObjectives
■■■ ProjectActivities
■■■ ProjectImpact
■■■ ProjectTimeline
■■■ ProjectStories
■■■ ProjectGallery
■■■ ProjectReports
■■■ ProjectPartners
■■■ GetInvolvedCTA
```

4. BACKEND — Administration & Content Management

The backend is the private system used by authorized Eco Warriors administrators and staff.

- Admin Dashboard
- Programs management
- Projects management
- Activities management
- Stories management
- Gallery/media management
- Reports/document management
- News management
- Events management
- Leadership/team management
- Partners management
- Impact statistics management
- Volunteer management
- Contact/message management
- Users and permissions
- System settings

5. Admin Dashboard

```
/admin
■■■ Dashboard
■■■ Programs
■■■ Projects
■■■ Activities
■■■ Stories
■■■ Gallery
■■■ Reports
■■■ News
■■■ Events
■■■ Partners
■■■ Team / Leadership
■■■ Volunteers
■■■ Impact Statistics
■■■ Messages
■■■ Users
```

6. Program Management

Program fields

- Program ID
- Name
- Slug
- Description
- Strategic Objective / Goal
- Program Image
- Status
- Projects
- Impact Metrics
- Created / Updated dates

Current strategic program areas

- Food Security & Sustainable Procurement
- Sexual & Reproductive Health and Rights
- Leadership & SDG Awareness
- Climate Action & Environmental Stewardship
- Digital Literacy & Entrepreneurship Skills
- Sport, Health & Community Development as a complementary website program area

7. Project Management

Projects are specific initiatives delivered under broader program areas.

Project fields

- Project ID
- Program ID
- Title
- Slug
- Short and full description
- Objectives
- Location
- Start / end dates
- Status
- Hero image
- Target beneficiaries

- Impact statistics
- Partners
- Created / updated dates

Example

Program: Sexual & Reproductive Health and Rights → Project: Kick Out Period Poverty

8. Activity Management

- Activity ID and Project ID
- Activity title
- Date and location
- Description and objectives
- Participants
- Results
- Photos / video
- Partners
- Related report

A project can contain many activities, for example education outreaches, reusable-pad distribution, football advocacy, male engagement and pad-production training.

9. Story Management

- Person name and profile
- Age / location where appropriate
- Photograph
- Headline
- Challenge
- Eco Warriors intervention
- Outcome / change
- Quote
- Full story
- Project relationship
- Draft / Published / Archived status

10. Gallery & Media Management

- Upload one or many images
- Connect images to a project and activity
- Captions
- Dates

- Optional location / photographer credit
- Image optimization
- Video or video-thumbnail support
- Full-screen frontend gallery

Recommended storage structure:

```
S3 / media
    ■■■ projects/
    ■ ■■■ kick-out-period-poverty/
    ■ ■■■ eco-digital/
    ■ ■■■ waste-to-wealth/
    ■ ■■■ my-period-my-tree/
    ■■■ leadership/
    ■■■ stories/
    ■■■ events/
    ■■■ gallery/
    ■■■ reports/
```

11. Reports Management

- Annual Impact Reports
- Strategic Plans
- Project Reports
- Evaluation / learning documents
- Research and publications
- PDF upload and download
- Report cover image and metadata

12. News & Events Management

News fields

- Title
- Slug
- Featured image
- Content
- Author
- Category
- Tags
- Publication date
- Status

Event fields

- Title
- Description
- Date

- Time
- Location
- Registration link
- Featured image
- Photos
- Results
- Report

13. Leadership & Partners Management

Leadership

- Name
- Position
- Biography
- Photo
- Social links
- Category
- Display order

Partners

- Organization name
- Logo
- Description
- Website
- Partnership type
- Related projects
- Display order

14. Impact Management

Impact numbers should be managed in the backend rather than hard-coded into the frontend.

- Metric title
- Number / value
- Program
- Project where applicable
- Year / reporting period
- Description
- Verification / source note
- Published status

Example: '140+ Youth & Women Trained' can be stored as an impact metric and displayed automatically on the homepage and impact page. The 2025 Impact Report records 140 youth and women completing the digital-literacy program.

15. Volunteer Management

- Volunteer applications
- Country and contact information
- Skills
- Area of interest
- Availability
- CV / supporting document
- Status: applicant, approved, active, alumni
- Skill-matching to program areas

The 2026 Strategic Plan calls for a formal database of national and international university volunteers for skill-matching across program pillars.

16. Contact & Message Management

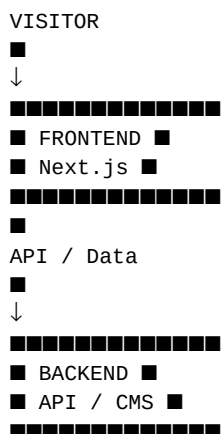
- Contact form submissions enter the admin dashboard.
- Administrators can mark messages as read, replied or archived.
- Separate enquiry categories can be used for partnerships, volunteering, media, donations and general enquiries.

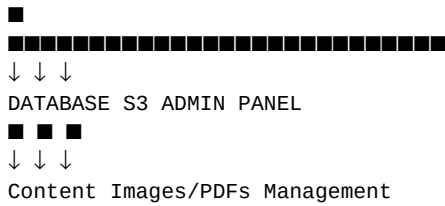
17. Database Relationship



This relationship lets the team add new content without redesigning the website.

18. Complete System Architecture





19. Recommended Technology Stack

Frontend

- Next.js
- TypeScript
- Tailwind CSS
- Responsive/mobile-first UI

Backend

- Option A: Next.js full-stack using API routes / Server Actions + PostgreSQL + S3.
- Option B: Separate Node.js / NestJS backend + REST API + PostgreSQL + S3.

Recommended direction

For a growing organization that may later need a mobile application, partner portal or other integrations, a separate backend API provides stronger long-term separation. For a simpler first release, Next.js full-stack is easier to operate.

20. Recommended Next.js Route Structure

```
app/
  page.tsx
  about/page.tsx
  about/leadership/page.tsx
  programs/page.tsx
  programs/[slug]/page.tsx
  projects/page.tsx
  projects/[slug]/page.tsx
  projects/[slug]/activities/page.tsx
  projects/[slug]/stories/page.tsx
  projects/[slug]/gallery/page.tsx
  projects/[slug]/reports/page.tsx
  impact/page.tsx
  stories/page.tsx
  news/page.tsx
  events/page.tsx
  gallery/page.tsx
  reports/page.tsx
  partners/page.tsx
  get-involved/page.tsx
  donate/page.tsx
  contact/page.tsx
```

21. Recommended First Release

- Home
- About and Leadership

- Programs and Program Details
- Projects and Project Details
- Kick Out Period Poverty project page
- Eco Digital project page
- Waste to Wealth project page
- My Period, My Tree project page
- Impact
- Stories
- Gallery
- Reports
- Partners
- Get Involved
- Donate
- Contact
- Admin dashboard with content management

22. Final Architecture Principle

FRONTEND = PUBLIC EXPERIENCE

BACKEND = CONTENT, DATA & ORGANIZATIONAL MANAGEMENT

DATABASE = STRUCTURED INFORMATION

AWS S3 = IMAGES, DOCUMENTS & MEDIA

API = CONNECTION BETWEEN FRONTEND AND BACKEND

Source note: The organizational terminology and strategic program structure in this blueprint follow the Eco Warriors Strategic Plan 2026. Impact examples should be published only after checking the corresponding Eco Warriors reports and project documentation.